SWATI AGGRAWAL

swatiagg357@gmail.com | +91-7011930926 | github.com/swatiaggrawal | linkedin.com/in/swatiaggrawal

PROFESSIONAL SUMMARY

M.Sc. Computer Science graduate (Sapienza Università di Roma) specializing in AI/ML, with an IEEE publication and thesis research applying **statistical benchmarking and comparative model evaluation** to Vision Transformers. Skilled in building and evaluating predictive models — from classical ML (Random Forest, SVM, Decision Trees) to deep learning (CNNs, Transformers) — and translating results into clear, stakeholder-ready findings. Hands-on across the full data science lifecycle: preprocessing, feature engineering, experimentation, evaluation, and production deployment (FastAPI, Docker, AWS). Proficient with AI-assisted development tools and Generative AI/LLM techniques (RAG, embeddings, prompt engineering).

SKILLS

Statistical & Predictive Modeling: Regression, Classification, Decision Trees, Random Forest, SVM, Clustering, Model Evaluation (Accuracy, Precision/Recall, F1, Confusion Matrix, Cross-Validation), GridSearchCV, Comparative Benchmarking
Deep Learning: CNNs, Vision Transformers, Transfer Learning, Continual Learning (EWC, Adapters), PyTorch, TensorFlow, Keras
Generative AI & LLMs: RAG, Prompt Engineering, Embeddings (Gemini API), Vector Databases (Pinecone), LLM-based Application Design
Data Science & Analytics: Pandas, NumPy, Matplotlib, Feature Engineering, EDA, Statistical Analysis, Data Preprocessing
Programming, Tools & Cloud: Python, SQL, C++, Bash, Git, Linux, AWS (ECS, Fargate, CloudWatch), Azure, Docker, FastAPI, REST APIs

EDUCATION

Sapienza Università di Roma — M.Sc. Computer Science **2026**
Relevant Coursework: Machine Learning, Deep Learning & Applied AI, Natural Language Processing, Computer Vision, Distributed Systems

- Thesis: Evaluating Pretrained Vision Transformers Under Continual Learning — designed and ran a **comparative benchmarking study** across three modeling approaches.

Sharda University — B.Tech, CSE with AI and ML **2023**
Relevant Coursework: Machine Learning, Data Structures, Databases, Operating Systems, Cloud Computing | Merit scholarship (Top 5–10% of cohort)

RESEARCH EXPERIENCE

Researcher, Vision Lab — Sapienza Università di Roma (Master's Thesis) **Nov 2025 – May 2026**
github.com/swatiaggrawal/Thesis-Vision-Transformers

- Designed a controlled experimental pipeline to **benchmark three modeling approaches** for continual learning — adapter-based fine-tuning + Elastic Weight Consolidation (EWC), naive Vision Transformer fine-tuning, and a ResNet-18 baseline — under identical evaluation protocol.
- Achieved **90.59% average accuracy with 2.01% average forgetting** on Split CIFAR-100 (5 sequential tasks) using the adapter+EWC approach, a ~97% relative reduction in catastrophic forgetting vs. the CNN baseline (24.22% avg. accuracy) and vs. naive ViT fine-tuning (41.91% avg. accuracy).
- Validated results via cross-seed evaluation to test robustness of reported metrics, and documented findings for reproducibility.

Machine Learning for Air Pollution — IEEE Conference Publication **2023**
doi.org/10.1109/ViTECoN58111.2023.10157028

- Ran a **comparative statistical evaluation of four modeling techniques** — Random Forest, SVM, Decision Tree, and Neural Network — on environmental air-quality datasets.
- Assessed models on accuracy, robustness, and generalization, and authored/presented the peer-reviewed findings.

PROJECTS

Metal Defect Detection — Comparative Modeling Study
github.com/swatiaggrawal/Metal-Defect-Detection

- Ran a **head-to-head comparison of three model families** — a custom CNN (TensorFlow/Keras), SVM, and Random Forest (HOG features, tuned via GridSearchCV) — for automated defect detection across 6 classes on the NEU industrial dataset.
- Achieved **97.22% test accuracy with the CNN vs. 86.11% for both SVM and Random Forest**, evaluated via accuracy, confusion matrices, precision, recall, and F1-score.

Knowledge Copilot — Grounded RAG Chatbot
github.com/swatiaggrawal/RAG-Chatbot | *Live demo: rag-chatbot-new.onrender.com*

- Built an end-to-end RAG pipeline (ingestion → embedding → retrieval → generation), integrating Gemini API embeddings with Pinecone; returns per-answer source citations to prevent hallucination.

DocDelta — Document Intelligence & Comparison Platform github.com/swatiaggrawal/DocDelta

- Built a semantic document-comparison pipeline (Sentence Transformers, cosine-similarity retrieval) classifying sentence-level changes at ~840ms warm-request latency; containerized and deployed on AWS ECS/Fargate with CloudWatch monitoring.